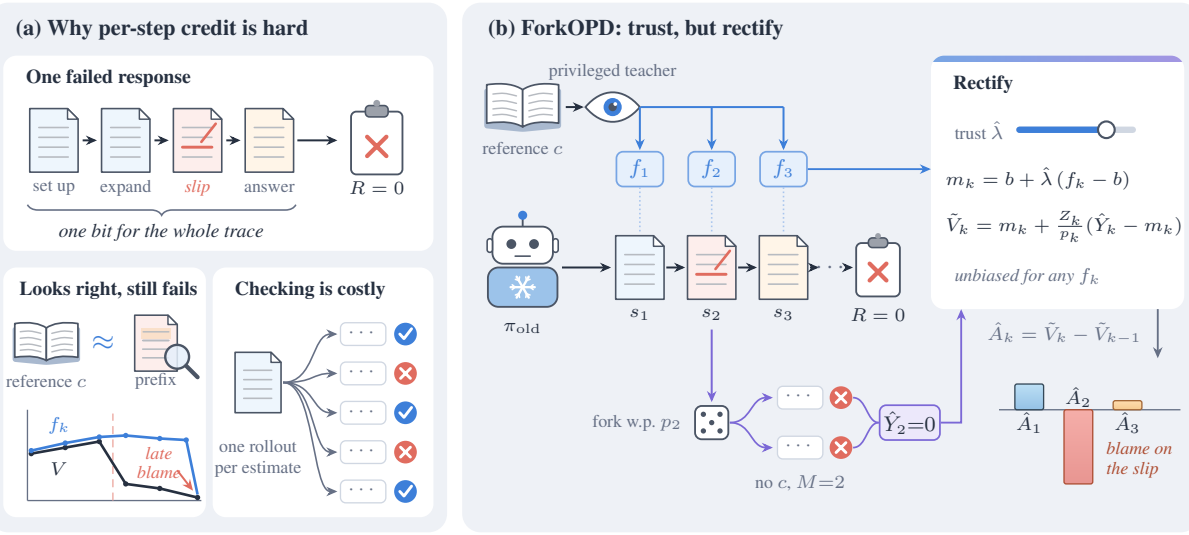

(a) Why per-step credit is hard
One failed response
set up
expand
slip
answer
R = 0
one bit for the whole trace
Looks right, still fails
reference c
prefix
f_k
V
late blame
Checking is costly
one rollout per estimate
(b) ForkOPD: trust, but rectify
privileged teacher
reference c
f_1
f_2
f_3
π_old
s_1
s_2
s_3
R = 0
fork w.p. p_2
no c, M=2
Ŷ_2=0
Rectify
trust λ̂
m_k = b + λ̂ (f_k − b)
V̂_k = m_k + (Z_k / p_k)(Ŷ_k − m_k)
unbiased for any f_k
Â_k = V̂_k − V̂_{k−1}
Â_1
Â_2
Â_3
blame on the slip